

The sharp Berry–Esseen bound for independent arrays with small Lyapunov ratio

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11 September 2026

Abstract

There is a universal number $\ell_0 > 0$ such that every finite independent centered array with positive total variance and finite third absolute moments satisfies $\Delta \leq C_E \ell$ whenever $\ell \leq \ell_0$, where $C_E = (3 + \sqrt{10})/(6\sqrt{2\pi})$. Here Δ is the Kolmogorov distance of the standardized sum from the standard normal law, and ℓ is its Lyapunov ratio. The constant is optimal. The proof adapts the uniform jitter, support contact, and local comparison arguments of He and Cheng to independent summands with different laws. A penalized maximization preserves small Lyapunov ratio. For the maximizing arrays, a Fourier expansion and the equality case of Esseen’s moment inequality determine a common scale. The variational equations confine each of their supports to two small clusters or to a small neighborhood of zero. An exact comparison with common-diameter Bernoulli arrays completes the proof. The latter bound is supplied by a companion paper and uses validated finite calculations. The present argument gives no numerical value of ℓ_0 .

1 The theorem and its inputs

Let $X_1, \dots, X_n$ be independent centered real random variables. Write

$$B^2 = \sum_{i=1}^n \mathbb{E} X_i^2 > 0, \quad T = \sum_{i=1}^n \mathbb{E} |X_i|^3 < \infty, \quad \ell = \frac{T}{B^3}, \quad (1.1)$$

and let

$$\Delta = \sup_{x \in \mathbb{R}} \left| \mathbb{P} \left\{ B^{-1} \sum_{i=1}^n X_i \leq x \right\} - \Phi(x) \right|, \quad \mathcal{R}(X_1, \dots, X_n) = \frac{\Delta}{\ell}.$$

Here Φ and ϕ are the standard normal distribution function and density. Zero summands are allowed. Set

$$c_E = 3 + \sqrt{10}, \quad C_E = \frac{c_E \phi(0)}{6}.$$

Theorem 1.1. *There exists an absolute constant $\ell_0 > 0$ such that, for every integer $n \geq 1$ and every array satisfying (1.1),*

$$\ell \leq \ell_0 \implies \Delta \leq C_E \ell.$$

The constant C_E cannot be decreased, even if all summands have the same two-point law.

The unrestricted conjecture $\Delta \leq C_E \ell$ remains open for general independent arrays. Theorem 1.1 excludes violations whose Lyapunov ratios tend to zero; its proof does not give an explicit threshold.

He and Cheng [3] prove the sharp bound for iid summands once the sample size exceeds a universal threshold. We use their uniform-jitter expansion and exact local comparison as the

starting points for the independent-array argument. Their contamination method also supplies the support equations used below. The replacements for identical distribution are proved here: selection by a smooth moment penalty, a variance-weighted Fourier construction, arithmetic restrictions on every support point, and conditional estimates for a weighted Poisson-binomial sum.

We record the two sharp inputs. Define

$$p_E = \frac{4 - \sqrt{10}}{2}, \quad q_E = 1 - p_E, \quad d_E = \sqrt{p_E q_E}, \quad \tau_E = p_E^2 + q_E^2. \quad (1.2)$$

Let Z_E have the centered variance-one law

$$\text{Law}(Z_E) = q_E \delta_{-p_E/d_E} + p_E \delta_{q_E/d_E}, \quad \beta_E = \mathbb{E}|Z_E|^3 = \frac{\tau_E}{d_E}, \quad \rho_E = \beta_E^{-1}.$$

Its third moment is positive.

Esseen's lattice moment theorem [2, pp. 161–162] has the following homogeneous form. If a centered random variable Y with finite third absolute moment is supported on a translate of $h\mathbb{Z}$, $h > 0$, then

$$|\mathbb{E}Y^3| + 3h\mathbb{E}Y^2 \leq c_E \mathbb{E}|Y|^3. \quad (1.3)$$

For positive variance, equality holds exactly for $\text{Law}(Y) = \text{Law}(\pm h d_E Z_E)$. In particular, the maximal span of an equality law is h . A constant centered variable has both sides zero. The corresponding iid asymptotic formula gives $\mathcal{R}(Z_E, \dots, Z_E) \rightarrow C_E$ as the number of summands tends to infinity [2, 4]. Its Lyapunov ratio is $\beta_E/\sqrt{n}$, proving the optimality assertion in Theorem 1.1.

The finite input is the common-diameter theorem of the companion paper [1, Theorem 1.2(ii)]:

$$\mathcal{R}(h(B_1 - p_1), \dots, h(B_m - p_m)) \leq C_E \quad (h > 0), \quad (1.4)$$

where the B_i are independent Bernoulli variables with arbitrary parameters $p_i \in [0, 1]$ and positive total variance. That theorem uses Schulz's binomial bound [5, Theorem 1] and the validated finite calculations described in the companion paper. The numerical dependency of Theorem 1.1 is through (1.4).

We also use the usual independent Berry–Esseen inequality with some finite universal constant C_U ; no particular numerical value is needed [6]. For a real random variable S , put $F_S(x) = \mathbb{P}(S \leq x)$ and $F_S^-(x) = \mathbb{P}(S < x)$. Since the normal distribution function is continuous,

$$\sup_x |F_S(x) - \Phi(x/B)| = \sup_x \max\{F_S(x) - \Phi(x/B), \Phi(x/B) - F_S^-(x)\} \quad (1.5)$$

when S is centered with variance B^2 . Reflection changes the second branch into the first. Constants denoted by C and c below may change between estimates. Subscripts specify fixed parameters on which they may depend. We write $A \asymp_\alpha B$ when $c_\alpha B \leq A \leq C_\alpha B$ for positive constants depending only on α . Limits of arrays always refer to finite rows; their lengths may grow without a prescribed rate.

The proof proceeds by contradiction. Section 2 selects exact maximizers whose Lyapunov ratios tend to zero and bounds each support at that scale. Sections 3 and 4 identify the common scale and confine the supports. Section 5 proves the exact local comparison needed for the final contradiction.

2 Selecting maximizers and bounding their supports

Suppose that Theorem 1.1 fails. Normalize each array to $B = 1$. For integers $j \geq 1$, let

$$\eta_j = j^{-1}, \quad t_j = e^{-j^2}, \quad f(r) = \begin{cases} 0, & r \leq 0, \\ r^2/(1+r), & r > 0, \end{cases} \quad \psi_j(t) = \eta_j f(\log(t/t_j)).$$

The function ψ_j is continuously differentiable and nondecreasing, and

$$\psi_j \geq 0, \quad \psi_j(t) = 0 \quad (t \leq t_j), \quad 0 \leq t\psi_j'(t) \leq \eta_j, \quad \psi_j(t) \geq \eta_j \{\log(t/t_j) - 1\}. \quad (2.1)$$

Choose a violating array with $\ell \leq t_j$, and let n_j be its length. In that fixed dimension maximize, over independent centered arrays of total variance one and thresholds $z \in \mathbb{R}$,

$$\frac{F_S(z) - \Phi(z)}{\ell} - \psi_j(\ell), \quad S = \sum_i X_i. \quad (2.2)$$

Its supremum exceeds C_E , since reflection permits a positive branch of the chosen violation.

Lemma 2.1. *The supremum in (2.2) is attained. For a maximizer, write*

$$R_j = \frac{F_{S_j}(z_j) - \Phi(z_j)}{\ell_j}.$$

Then

$$R_j > C_E, \quad R_j \leq C_U, \quad \ell_j \leq t_j \exp\{1 + (C_U - C_E)/\eta_j\} \longrightarrow 0. \quad (2.3)$$

Moreover, $\mathcal{R}(X_{1,j}, \dots, X_{n_j,j}) = R_j$.

Proof. For a fixed dimension n , Lyapunov's inequality and convexity give

$$\ell \geq \sum_i (\mathbb{E}X_i^2)^{3/2} \geq n^{-1/2}.$$

Along a maximizing sequence with objective greater than a fixed $r_0 > 0$, the positive discrepancy is at least $r_0\ell$ and at most one, so $\ell \leq r_0^{-1}$. Each coordinate has a bounded third absolute moment. Coordinatewise weak subsequences preserve means and variances, because this bound gives uniform integrability of squares. The limiting total variance is therefore one.

The thresholds stay bounded. Uniformly over centered variance-one sums, Chebyshev's inequality makes their positive discrepancies tend to zero as the threshold tends to either infinity, whereas here they are at least $r_0/\sqrt{n}$. Pass to a subsequence on which the thresholds converge to z and the third-moment sums converge to L . Moving-threshold Portmanteau gives $F_S(z) \geq \limsup F_{S_r}(z_r)$, and lower semicontinuity gives the limiting third-moment sum $\ell \leq L$. If $\ell < L$, the limiting positive ratio strictly increases and its nondecreasing penalty weakly decreases, exceeding the supremum. Thus $\ell = L$ and the limit attains the supremum. This is the fixed-dimension compactness argument of He and Cheng [3, Lemma 4.1], applied coordinatewise.

At the maximizer, $R_j - \psi_j(\ell_j) > C_E$ and $R_j \leq C_U$. Equation (2.1) gives (2.3). A larger discrepancy of either sign in the same array, with reflection if necessary, would increase (2.2); hence its full ratio is R_j . $\square$

We suppress j temporarily. Put

$$s_i = \mathbb{E}X_i^2, \quad b_i = \mathbb{E}|X_i|^3, \quad M_i = \mathbb{E}[X_i|X_i|], \quad K = R + \ell\psi_j'(\ell).$$

Thus $R \leq K \leq R + \eta_j$. Let $F = F_S$ and let G_i be the CDF of $S - X_i$.

Lemma 2.2 (Support contact). *For every $y \in \mathbb{R}$,*

$$\begin{aligned} D_i(y) &:= G_i(z - y) - F(z) + \phi(z)y + \frac{z\phi(z)}{2}(y^2 - s_i) \\ &\quad + \frac{3}{2}K\ell(y^2 - s_i) - K\{|y|^3 - b_i - 3M_i y\} \leq 0. \end{aligned} \quad (2.4)$$

Equality holds at every $y \in \text{supp } X_i$.

Proof. Replace the raw law of coordinate i by $(1 - e) \text{Law}(X_i) + e\delta_y$, keeping the raw threshold z fixed, and then center and standardize the resulting array. The derivatives at $e = 0$ of its total mean, total variance, and sum of absolute third central moments are, respectively,

$$y, \quad y^2 - s_i, \quad |y|^3 - b_i - 3M_i y.$$

For the third identity, differentiation under the expectation is justified by a bound $C_y(1 + X_i^2)$. The raw CDF derivative is $G_i(z - y) - F(z)$. Consequently the derivative of the normalized third-moment sum is

$$|y|^3 - b_i - 3M_i y - \frac{3}{2}\ell(y^2 - s_i).$$

Differentiating (2.2) and multiplying by ℓ gives (2.4). Maximality makes it nonpositive.

Its integral under $\text{Law}(X_i)$ is zero: each displayed moment term integrates to zero, and $\mathbb{E}G_i(z - X_i) = F(z)$. Thus $D_i = 0$ almost surely. A CDF is upper semicontinuous, so D_i is upper semicontinuous. The closed set $\{D_i = 0\}$ therefore contains the entire support. This is the independent-coordinate version of He and Cheng [3, Lemma 4.3]; the moment coefficient K includes the derivative of the penalty. $\square$

For $0 \leq s \leq 1/2$ define the Gaussian remainder

$$H_s(z, y) = \Phi\left(\frac{z - y}{\sqrt{1 - s}}\right) - \Phi(z) + \phi(z)y + \frac{z\phi(z)}{2}(y^2 - s).$$

The cancellation used in He and Cheng [3, Lemma 4.4] gives

$$|H_s(z, y)| \leq \frac{\phi(0)}{6}|y|^3 + C(s|y| + s^2), \quad (2.5)$$

$$H_s(z, y) = \phi''(z) \left(\frac{sy}{2} - \frac{y^3}{6} \right) + O(y^4 + sy^2 + s^2) \quad (2.6)$$

in the second formula for bounded y , uniformly in z . For completeness, translation Taylor expansion has remainder at most $\|\phi''\|_\infty |y|^3/6 = \phi(0)|y|^3/6$. For $u = z - y$, differentiating $\lambda \mapsto \Phi(u/\sqrt{1 - \lambda})$ twice shows that its second derivative is uniformly bounded for $0 \leq \lambda \leq 1/2$. The scale correction, after cancellation of the term $sz\phi(z)/2$, is

$$\frac{s}{2}\{(z - y)\phi(z - y) - z\phi(z)\} + O(s^2).$$

The derivative of $z\phi(z)$ is $-\phi''(z)$ and all the Gaussian derivatives in question are bounded. These facts give both formulas.

Delete coordinate i , and suppose $s_i \leq 1/2$. Its Lyapunov ratio is

$$\ell_- = \frac{\ell - b_i}{(1 - s_i)^{3/2}}.$$

Adding a zero coordinate puts the deleted array in the same maximization class. Its positive signed ratio at every threshold is therefore at most $R + \psi_j(\ell_-) - \psi_j(\ell)$. With $\log_+ t = \max(\log t, 0)$, equation (2.1) gives

$$\psi_j(\ell_-) - \psi_j(\ell) \leq \eta_j \log_+(\ell_-/\ell) \leq 3\eta_j s_i.$$

Substitution into the contact equality yields, for $y \in \text{supp } X_i$,

$$\begin{aligned} 0 \leq H_{s_i}(z, y) - K|y|^3 + \frac{3}{2}K\ell y^2 + 3KM_i y \\ + C\ell s_i^2 + \eta_j b_i + C\eta_j \ell s_i. \end{aligned} \quad (2.7)$$

Here and below these estimates are used for large j , so all the constants are uniform. To verify the remainder, the constant terms before applying upper bounds are

$$R(\ell_- - \ell) - \frac{3}{2}K\ell s_i + Kb_i + \{\psi_j(\ell_-) - \psi_j(\ell)\}\ell_-.$$

Writing $K = R + \vartheta$, $0 \leq \vartheta \leq \eta_j$, the terms with R equal

$$R\ell\{(1 - s_i)^{-3/2} - 1 - \frac{3}{2}s_i\} - Rb_i\{(1 - s_i)^{-3/2} - 1\}.$$

The second is nonpositive. The remaining terms are bounded above by those in (2.7).

The support estimate below adapts the cubic domination argument of He and Cheng [3, Proposition 4.5] to the coordinate variance scales.

Proposition 2.3. *There is a finite constant L such that the selected arrays satisfy*

$$\text{supp } X_{i,j} \subset [-L\ell_j, L\ell_j] \quad \text{for all } i \text{ and all sufficiently large } j. \quad (2.8)$$

Proof. The contact equality first bounds every support by one fixed interval: its cubic coefficient is at most $-C_E$, while all coefficients of lower degree are uniformly bounded. Indeed $\ell \leq 1/C_E$, $b_i \leq \ell$, $s_i \leq 1$, $|M_i| \leq s_i$, and $\phi(z)$, $z\phi(z)$, R , K are bounded. Thus each support is compact. Let $L_i = \max\{|y| : y \in \text{supp } X_i\}$. Since $s_i^{3/2} \leq b_i \leq \ell_j \rightarrow 0$, eventually every $s_i \leq 1/2$. Also $b_i \leq s_i L_i$.

Use (2.5) in (2.7) at an extreme support point. Since $\gamma := C_E - \phi(0)/6 > 0$,

$$\gamma L_i^3 \leq C\{\ell L_i^2 + s_i L_i + s_i^2 + \eta_j \ell s_i\}.$$

If L_i is a sufficiently large multiple of $\ell + \sqrt{s_i}$, division by L_i^3 contradicts this inequality. Hence

$$L_i \leq C(\ell + \sqrt{s_i}). \quad (2.9)$$

Suppose now that some sequence of coordinates has $\sigma_j/\ell_j \rightarrow \infty$, where $\sigma_j = \sqrt{s_{i,j}}$. Then $\sigma_j \rightarrow 0$, and (2.9) bounds the centered variance-one laws $X_{i,j}/\sigma_j$ in a common compact interval. Extract a weak limit Q , still centered with variance one, and further subsequences such that

$$R_j \rightarrow r \geq C_E, \quad \frac{\phi''(z_j)}{6} \rightarrow d, \quad \frac{M_{i,j}}{\sigma_j^2} \rightarrow m, \quad |d| \leq \phi(0)/6.$$

For any $u \in \text{supp } Q$ choose support points $\sigma_j u_j$ with $u_j \rightarrow u$. Divide (2.7) by σ_j^3 and use (2.6). The quadratic term tends to zero. The remainders tend to zero because

$$\frac{\ell s_i^2}{\sigma_j^3} = \ell_j \sigma_j \rightarrow 0, \quad \frac{\eta_j b_i}{\sigma_j^3} \leq C\eta_j(1 + \ell_j/\sigma_j) \rightarrow 0, \quad \frac{\eta_j \ell s_i}{\sigma_j^3} = \eta_j \ell_j / \sigma_j \rightarrow 0.$$

We obtain

$$0 \leq -du^3 - r|u|^3 + Au, \quad A = 3(d + rm).$$

The support of Q contains $u_+ > 0$ and $u_- < 0$. The first requires $A \geq (r + d)u_+^2 > 0$, while the second requires $A \leq -(r - d)|u_-|^2 < 0$. This contradiction proves $\sup_i \sigma_{i,j} = O(\ell_j)$, and (2.9) proves (2.8). $\square$

3 Uniform jitter and equality in the moment bound

We give the Fourier argument for arbitrary independent rows satisfying

$$\sum_i \mathbb{E} X_{i,j}^2 = 1, \quad \ell_j = \sum_i \mathbb{E} |X_{i,j}|^3 \longrightarrow 0, \quad |X_{i,j}| \leq L\ell_j.$$

Put $Y_{i,j} = X_{i,j}/\ell_j$ and take independent copies $Y'_{i,j}$. The measures

$$\mu_j(A) = \frac{\ell_j^2}{2} \sum_i \mathbb{E} [(Y_{i,j} - Y'_{i,j})^2 \mathbf{1}_{\{Y_{i,j} - Y'_{i,j} \in A\}}] \quad (3.1)$$

are probability measures supported in $[-2L, 2L]$. Pass to a subsequence on which $\mu_j \Rightarrow \mu$ and

$$\theta_j = \ell_j^{-1} \sum_i \mathbb{E} X_{i,j}^3 \longrightarrow \theta, \quad |\theta| \leq 1.$$

Write $f_{i,j}(u) = \mathbb{E} e^{iuY_{i,j}}$ and

$$g_j(u) = \ell_j^2 \sum_i (1 - |f_{i,j}(u)|^2) = 2 \int \frac{1 - \cos(ud)}{d^2} \mu_j(dd) \longrightarrow g(u) = 2 \int \frac{1 - \cos(ud)}{d^2} \mu(dd). \quad (3.2)$$

The quotient at $d = 0$ is $u^2/2$. This convergence holds locally in C^2 . A nonzero zero u of g requires $\mu(\{0\}) = 0$ and $ud \in 2\pi\mathbb{Z}$ for μ -almost every d . The zero set is consequently a closed additive subgroup. Since $g''(0) = 2$, zero is isolated, and the subgroup is either $\{0\}$ or $(2\pi/h)\mathbb{Z}$ for some $h > 0$. Put $h = 0$ in the first case. At every zero of g , its second derivative is two.

Lemma 3.1 (Uniform-jitter expansion). *Let V_h be independent and uniform on $[-h/2, h/2]$, with $V_0 = 0$, and put $\kappa_j = \sum_i \mathbb{E} X_{i,j}^3$. Then*

$$\sup_x \left| \mathbb{P} \left\{ \sum_i X_{i,j} + \ell_j V_h \leq x \right\} - \Phi(x) - \frac{\kappa_j}{6} (1 - x^2) \phi(x) \right| = o(\ell_j). \quad (3.3)$$

In particular, uniformly in x ,

$$F_{S_j}(x) - \Phi(x) \leq \ell_j \phi(x) \left\{ \frac{h}{2} + \frac{\theta_j}{6} (1 - x^2) \right\} + o(\ell_j), \quad (3.4)$$

$$\Phi(x) - F_{S_j}^-(x) \leq \ell_j \phi(x) \left\{ \frac{h}{2} - \frac{\theta_j}{6} (1 - x^2) \right\} + o(\ell_j). \quad (3.5)$$

Proof. We use the signed smoothing inequality of He and Cheng [3, Lemma 2.1]. In its needed form, if F is a probability CDF with finite first absolute moment, and ν is a finite signed measure of mass one with $\int |x| d|\nu|(x) < \infty$ whose CDF G is globally Lipschitz with constant M , then

$$\|F - G\|_\infty \leq C_1 \int_{-A}^A \frac{|\widehat{F}(t) - \widehat{\nu}(t)|}{|t|} dt + \frac{C_2 M}{A} \quad (A > 0). \quad (3.6)$$

Our Fourier convention is $\widehat{\nu}(t) = \int e^{itx} \nu(dx)$, and $\text{sinc } u = \sin(u)/u$, with $\text{sinc } 0 = 1$. The signed CDF $G_j(x) = \Phi(x) + \kappa_j(1 - x^2)\phi(x)/6$ satisfies these hypotheses with a uniform Lipschitz constant, and

$$\widehat{\nu}_j(t) = e^{-t^2/2} \{1 + \kappa_j(it)^3/6\}.$$

The product bound replacing the iid power is

$$\left| \prod_i f_{i,j}(u) \right| \leq \exp\{-g_j(u)/(2\ell_j^2)\}. \quad (3.7)$$

At low standardized frequencies, $\sum_i \mathbb{E}X_{i,j}^4 \leq L\ell_j^2$ and $\sum_i (\mathbb{E}X_{i,j}^2)^2 \leq L^2\ell_j^2$. Taylor expansion of the logarithms, for $|t|\ell_j \leq \delta_L$ with $\delta_L > 0$ sufficiently small, gives

$$\sum_i \log \mathbb{E}e^{itX_{i,j}} = -\frac{t^2}{2} + \frac{(it)^3 \kappa_j}{6} + O_L(t^4 \ell_j^2).$$

The real part of the remainder is at most a fixed fraction of t^2 . Since the cubic term is purely imaginary, the inequalities $|e^{iv} - 1 - iv| \leq v^2/2$ for real v and $|e^w - 1| \leq |w|e^{|w|}$ imply

$$\left| \prod_i \mathbb{E}e^{itX_{i,j}} - \hat{\nu}_j(t) \right| \leq C_L e^{-ct^2} \ell_j^2 (t^4 + t^6) \quad (|t|\ell_j \leq \delta_L).$$

Multiplication by $\text{sinc}(h\ell_j t/2)$ adds an integrable $O_h(\ell_j^2 t^2 e^{-ct^2})$ term. Thus the low-frequency part of the integral in (3.6) is $O_L(\ell_j^2)$.

Fix a raw cutoff $T > \delta_L$ whose endpoints are not zeros of g , and set $A = T/\ell_j$. Near each nonzero zero $u_k = 2\pi k/h$ in $[-T, T]$, local C^2 convergence makes g_j strictly convex. Its unique nearby minimum $v_{j,k}$ tends to u_k , and nonnegativity gives $g_j(u) \geq (u - v_{j,k})^2/2$ on a fixed small neighborhood. The jitter multiplier vanishes at u_k , and is bounded there by $C(|u - v_{j,k}| + |v_{j,k} - u_k|)$. Since $|u|$ is bounded away from zero in this neighborhood, its contribution to the smoothing integral is at most

$$C \int_{\mathbb{R}} (|u - v_{j,k}| + |v_{j,k} - u_k|) e^{-(u - v_{j,k})^2/(4\ell_j^2)} du = O(\ell_j^2) + o(\ell_j).$$

On the remaining compact raw-frequency set g has a positive minimum, and (3.7) gives exponential decay. If $h = 0$ there are no nonzero zeros to consider. The Gaussian comparison transform is itself exponentially small at these frequencies. Equation (3.6) leaves $C\ell_j/T$. Taking the row limit first and then $T \rightarrow \infty$ proves (3.3). This is the displaced-resonance argument of He and Cheng [3, Lemma 2.2], with (3.1) controlling the independent product.

Let J_j be the jittered CDF. Bounded support of the jitter gives $J_j(x - h\ell_j/2) \leq F_{S_j}^-(x) \leq F_{S_j}(x) \leq J_j(x + h\ell_j/2)$ when $h > 0$. Taylor expansion gives (3.4)–(3.5). For $h = 0$, (3.3) also holds for left limits by continuity of G_j . This is the envelope step of He and Cheng [3, Corollary 2.4]. $\square$

Proposition 3.2 (Equality in the array bound). *For the selected arrays of Section 2,*

$$R_j \longrightarrow C_E, \quad z_j \longrightarrow 0, \quad h = \frac{1}{\tau_E} = \frac{c_E}{\sqrt{10}}, \quad \theta = \frac{1}{\sqrt{10}} \quad (3.8)$$

on every subsequence chosen above. In addition,

$$\sum_i \mathbb{E}X_{i,j}^2 \delta_{\text{Law}(X_{i,j}/\ell_j)} \implies \delta_{\text{Law}(\rho_E Z_E)}. \quad (3.9)$$

The outer measures in (3.9) are probability measures on the space of laws supported in $[-L, L]$.

Proof. The envelopes give

$$\limsup_j R_j \leq \frac{\phi(0)}{6} (|\theta| + 3h). \quad (3.10)$$

If $h = 0$, the right side is at most $\phi(0)/6 < C_E$, which is impossible. Suppose $h > 0$ and put $u_0 = 2\pi/h$. For each i, j , choose the phase of $f_{i,j}(u_0)$, arbitrarily when that value is zero, and round $Y_{i,j}$ to the nearest point on the corresponding translate of $h\mathbb{Z}$. Call the rounded variable $Z_{i,j}$ and its centered version $Q_{i,j}$. If d is the distance to the selected lattice, $|d| \leq h/2$, then $|e^{iu_0 Y_{i,j}} - e^{i \arg f_{i,j}(u_0)}| \geq 4|d|/h$. It follows that

$$\mathbb{E}|Y_{i,j} - Q_{i,j}|^2 \leq \mathbb{E}|Y_{i,j} - Z_{i,j}|^2 \leq \frac{h^2}{8}(1 - |f_{i,j}(u_0)|^2).$$

Consequently

$$e_j := \ell_j^2 \sum_i \mathbb{E}|Y_{i,j} - Q_{i,j}|^2 \longrightarrow 0. \quad (3.11)$$

All centered rounded variables lie in a fixed compact interval. For variables Y, Q in that interval, the differences of their second moments, signed third moments, and absolute third moments are bounded by a constant times $\mathbb{E}[|Y - Q|(|Y| + |Q|)]$. Summing by Cauchy–Schwarz and using (3.11) gives

$$\ell_j^2 \sum_i \mathbb{E}Q_{i,j}^2 \rightarrow 1, \quad \ell_j^2 \sum_i \mathbb{E}|Q_{i,j}|^3 \rightarrow 1, \quad \ell_j^2 \sum_i \mathbb{E}Q_{i,j}^3 \rightarrow \theta. \quad (3.12)$$

The second moments in the Cauchy–Schwarz bound have bounded sum: $\ell_j^2 \sum_i \mathbb{E}(|Y_{i,j}| + |Q_{i,j}|)^2 \leq 6 + 4e_j$. These estimates impose no bound on $n_j \ell_j^2$.

Apply (1.3) to each $Q_{i,j}$ and sum. Equations (3.12) imply

$$|\theta| + 3h \leq c_E.$$

Since $R_j > C_E$, equality holds here and in (3.10), and $R_j \rightarrow C_E$. In particular, both nonnegative quantities

$$\begin{aligned} & \ell_j^2 \sum_i \{c_E \mathbb{E}|Q_{i,j}|^3 - |\mathbb{E}Q_{i,j}^3| - 3h \mathbb{E}Q_{i,j}^2\}, \\ & \ell_j^2 \left\{ \sum_i |\mathbb{E}Q_{i,j}^3| - \left| \sum_i \mathbb{E}Q_{i,j}^3 \right| \right\} \end{aligned}$$

tend to zero.

Consider the compact class of centered laws on translates of $h\mathbb{Z}$ in a fixed bounded interval. For such a law Q write

$$v(Q) = \int y^2 Q(dy), \quad d(Q) = c_E \int |y|^3 Q(dy) - \left| \int y^3 Q(dy) \right| - 3hv(Q).$$

On subsets where $v(Q)$ is bounded below, continuity and Esseen's equality case imply that $d(Q)/v(Q)$ is bounded below by a positive number away from the two laws $\text{Law}(\pm h d_E Z_E)$. The same conclusion holds near variance zero. Indeed, if nondegenerate laws Q_r tend to δ_0 , its mass tends to one at a unique lattice site $c_r \rightarrow 0$. Represent a variable of law Q_r as $V_r = c_r + D_r$, where $D_r \in h\mathbb{Z}$, and let $a_r = \mathbb{P}(D_r \neq 0) \rightarrow 0$. Bounded support and centering give $|c_r| = O(a_r)$ and $\mathbb{E}D_r^2 \geq h^2 a_r$. Hence

$$\text{Var } V_r = \mathbb{E}D_r^2 \{1 + O(a_r)\}, \quad \frac{\mathbb{E}|V_r|^3}{\text{Var } V_r} \geq h - o(1).$$

Using $|\mathbb{E}V_r^3| \leq \mathbb{E}|V_r|^3$, we obtain

$$\frac{d(Q_r)}{v(Q_r)} \geq (c_E - 4)h - o(1) = (\sqrt{10} - 1)h - o(1) > 0. \quad (3.13)$$

The vanishing first deficit therefore concentrates variance weight on $\text{Law}(\pm h d_E Z_E)$. At these two laws, the signed third moment divided by variance is the nonzero number $\pm h(q_E - p_E)$. The vanishing second deficit forces one of the two orientation weights to tend to zero. The ratio of the aggregate absolute third moment to variance in (3.12) is one, whereas at either equality law it is $h d_E \beta_E = h \tau_E$. Thus $h = 1/\tau_E$ and the magnitude of θ is $h(q_E - p_E) = 1/\sqrt{10}$.

The positive branch determines its sign. Write $A = h/2$ and $D = |\theta|/6$. The possible positive envelopes are $E_\pm(x) = \phi(x)(A \pm D \mp Dx^2)$. Since $A > 3D$,

$$E'_-(x) = x\phi(x)(3D - A - Dx^2) < 0 \quad (x > 0),$$

so its maximum is $\phi(0)(A - D) < C_E$. The negative orientation cannot support $R_j > C_E$. The function E_+ has its unique global maximum $\phi(0)(A + D) = C_E$ at zero, by $\phi(x) \leq \phi(0)$ and $(1 - x^2)\phi(x) \leq \phi(0)$. Both envelopes tend to zero at infinity. Uniformity in (3.4) now gives $z_j \rightarrow 0$ and $\theta > 0$. This is the active-branch argument used in He and Cheng [3, Lemma 5.1].

Finally, (3.11) transfers the conclusion from rounded laws and their weights to the original ones. Explicitly,

$$\ell_j^2 \sum_i |\text{Var } Y_{i,j} - \text{Var } Q_{i,j}| \leq e_j^{1/2} \left(\ell_j^2 \sum_i \mathbb{E}(|Y_{i,j}| + |Q_{i,j}|)^2 \right)^{1/2} \rightarrow 0.$$

If $W_1(P, Q)$ denotes the infimum of $\mathbb{E}|Y - Z|$ over couplings with laws P, Q , the same coupling gives

$$\ell_j^2 \sum_i \text{Var}(Y_{i,j}) W_1(\text{Law}(Y_{i,j}), \text{Law}(Q_{i,j}))^2 \leq L^2 e_j \rightarrow 0.$$

On the compact space of bounded-support laws, this proves (3.9). Every subsequence admits the preceding argument with the same scale and positive orientation, so the assertions hold for the selected sequence. $\square$

4 Confinement of every support

Throughout this section we use the selected arrays, the scale $h = 1/\tau_E$, and the limits in Proposition 3.2. We first show that two support points of the same coordinate can have only an asymptotically integral separation in units of $h\ell_j$. This extends the support-separation argument of He and Cheng [3, Lemma 5.3].

Lemma 4.1 (Arithmetic support differences). *Choose arbitrary coordinate indices i_j and support points $x_j, y_j \in \text{supp } X_{i_j,j}$. Every subsequential limit of $(y_j - x_j)/\ell_j$ belongs to $h\mathbb{Z}$.*

Proof. Deleting any coordinate changes g_j and its first two derivatives on a fixed compact interval by $O(\ell_j^2)$, uniformly over that coordinate. Also, by Proposition 2.3,

$$s_i = O(\ell_j^2), \quad b_i = O(\ell_j^3), \quad |M_i| = O(\ell_j^2), \quad \ell_- = \ell_j + O(\ell_j^3).$$

The deleted arrays therefore have the same limiting function g . Lemma 3.1 applies uniformly to their standardized laws. Indeed, failure of uniformity would select one failing coordinate per row and contradict that lemma for the resulting sequence. We may use raw jitter width exactly $h\ell_j$: the natural width of a standardized deleted array, after returning to raw units, differs from it by $O(\ell_j^3)$. Coupling the two uniform jitters and using the bounded derivative of their smooth comparison CDF gives an $o(\ell_j)$ error.

At a support point $y = O(\ell_j)$, the contact equality gives

$$G_i(z_j - y) - F(z_j) = -\phi(z_j)y - \frac{z_j\phi(z_j)}{2}(y^2 - s_i) + O(\ell_j^3).$$

The Gaussian expansion has the same first two terms. Thus

$$G_i(z_j - y) - \Phi\left(\frac{z_j - y}{\sqrt{1 - s_i}}\right) = R_j\ell_j + O(\ell_j^3), \quad (4.1)$$

uniformly over all coordinates and their support points.

Fix the chosen coordinate i_j , put $T_j = S_j - X_{i_j,j}$ and $u_j = z_j - x_j$, and define locally finite measures

$$\nu_j(A) = \ell_j^{-1} \mathbb{P}\{(T_j - u_j)/\ell_j \in A\}.$$

Let J_j be the CDF of $T_j + \ell_j V_h$. The jitter envelopes bound ν_j on every compact interval: for fixed $a < b$, the mass on $[a, b]$ is at most

$$\ell_j^{-1} \{J_j(u_j + \ell_j(b + h/2)) - J_j(u_j + \ell_j(a - h/2))\}.$$

Using slightly enlarged endpoints gives the same bound if needed. The jitter expansion makes this bounded, so a subsequence has a vague limit ν . Since $u_j \rightarrow 0$, that expansion also gives

$$\ell_j^{-1} \{J_j(u_j + \ell_j b) - J_j(u_j + \ell_j a)\} \longrightarrow \phi(0)(b - a).$$

Convolution with a compactly supported measure passes to vague limits when tested against continuous functions of compact support. It follows that

$$\nu * \text{Unif}[-h/2, h/2] = \phi(0) dx. \quad (4.2)$$

Differentiation in the sense of distributions gives equality of the translates of ν by $h/2$ and $-h/2$. Hence ν is h -periodic.

Equations (4.1) and (3.8) give $J_j(u_j + h\ell_j/2) - G_{i_j}(u_j) = o(\ell_j)$. The exact identity

$$J_j(u_j + h\ell_j/2) - G_{i_j}(u_j) = \mathbb{E} \left[\left(1 - \frac{T_j - u_j}{h\ell_j} \right) \mathbf{1}_{\{u_j < T_j < u_j + h\ell_j\}} \right]$$

has a nonnegative integrand. After division by ℓ_j , it forces the limiting mass of each compact subinterval of $(0, h)$ to be zero. Therefore $\nu((0, h)) = 0$. Periodicity and (4.2) now identify

$$\nu = \phi(0)h \sum_{k \in \mathbb{Z}} \delta_{kh}. \quad (4.3)$$

Suppose $(y_j - x_j)/\ell_j \rightarrow d$. The second support contact is at $z_j - y_j = u_j - \ell_j d_j$. Its saturation, by the same nonnegative integral, gives no limiting mass in $(-d, h - d)$. If $d \notin h\mathbb{Z}$, one atom of (4.3) lies strictly inside that interval. A compact neighborhood of that atom contradicts saturation. Thus $d \in h\mathbb{Z}$. $\square$

Lemma 4.2 (Limits of entire coordinate supports). *Choose any coordinate i_j from each row, and pass to a subsequence on which $\text{Law}(X_{i_j,j}/\ell_j)$ converges weakly to Q and its support converges in Hausdorff distance to a nonempty compact set A . Then either $A = \{0\}$ and $Q = \delta_0$, or A consists of exactly two points at distance h and Q gives positive mass to both.*

Proof. All laws and supports lie in the fixed interval $[-L, L]$, so these subsequences exist. Put $v = \int y^2 Q(dy)$ and $M = \int y|y|Q(dy)$. For every $t \in A$, choose support points $y_j = \ell_j t_j$ with $t_j \rightarrow t$. In (2.7), set $s_{i,j} = \ell_j^2 v_j$ and $M_{i,j} = \ell_j^2 M_j$. Formula (2.6) is uniform for bounded t_j, v_j , including $v_j \rightarrow 0$. Divide by $\phi(0)\ell_j^3/6$ and use $R_j, K_j \rightarrow C_E$ and $z_j \rightarrow 0$. The remainder in (2.7), divided by ℓ_j^3 , is $O(\eta_j) + o(1)$. We obtain

$$t^3 - c_E |t|^3 + \frac{3}{2} c_E t^2 + 3(c_E M - v)t \geq 0 \quad (t \in A). \quad (4.4)$$

This is the independent-coordinate counterpart of the limiting support polynomial in He and Cheng [3, Lemma 5.2].

If Q is nondegenerate, the extreme points of A can be written $-a, b$ with $a, b > 0$, because $\text{supp } Q \subset A$ and Q is centered. Evaluating (4.4) at these two points and eliminating its linear coefficient gives

$$(c_E + 1)a^2 + (c_E - 1)b^2 \leq \frac{3}{2} c_E (a + b).$$

At fixed $a + b$, the left side is at least $(c_E^2 - 1)(a + b)^2/(2c_E)$. Since $c_E^2 - 1 = 6c_E$,

$$a + b \leq \frac{3c_E^2}{c_E^2 - 1} = \frac{c_E}{2} < 2h. \quad (4.5)$$

Lemma 4.1 says that every difference between points of A belongs to $h\mathbb{Z}$. Hence A has exactly two points and their distance is h . The limiting law has positive variance, so both masses are positive.

If Q is degenerate, centering makes it δ_0 , and $v = M = 0$. Equation (4.4) confines A to

$$\left[-\frac{3c_E}{2(c_E + 1)}, \frac{3c_E}{2(c_E - 1)} \right] \subset (-h, h).$$

Since $0 \in A$, Lemma 4.1 forces $A = \{0\}$. In particular, support points whose probabilities vanish in the weak limit are also covered. $\square$

Proposition 4.3 (Two clusters and the remaining coordinates). *For all sufficiently large j , partition the coordinates according to whether the support diameter of $Y_{i,j} = X_{i,j}/\ell_j$ is at least $h/2$. Let m_j count the coordinates in that class. There is an $\alpha \in (0, 1/2)$ for which those coordinates have representations*

$$Y_{i,j} = h_{i,j}(B_{i,j} - p_{i,j}) + U_{i,j}, \quad p_{i,j} \in [\alpha, 1 - \alpha], \quad \mathbb{E}(U_{i,j} \mid B_{i,j}) = 0, \quad (4.6)$$

with independent pairs, such that

$$\max_i |h_{i,j} - h| \rightarrow 0, \quad \max_i \|U_{i,j}\|_\infty \rightarrow 0.$$

The other coordinates, denoted $Z_{r,j}$ in these same scaled units, are centered, independent of all those pairs, and satisfy

$$\max_r \|Z_{r,j}\|_\infty \rightarrow 0, \quad m_j \asymp \ell_j^{-2} \rightarrow \infty, \quad \sum_r \text{Var } Z_{r,j} = o(m_j). \quad (4.7)$$

Proof. Lemma 4.2 and compactness make the following conclusions uniform over coordinates. Supports with diameter below $h/2$ have diameters tending to zero. The other diameters tend to h , and their supports approach their two extreme points. Otherwise one could select a coordinate per row whose simultaneous law and support limit violated that lemma.

Divide each such support at the midpoint of its extreme points, and let $B_{i,j}$ indicate the upper cluster. Both cluster probabilities are bounded away from zero uniformly. If one tended to zero, the weak limit would be a point mass, while the full support limit would retain two points at distance h , contrary to Lemma 4.2. Choose $\alpha > 0$ accordingly. The difference of the two conditional means is $h_{i,j} \rightarrow h$ uniformly. Subtracting these means defines $U_{i,j}$ and gives (4.6); its supremum norm tends uniformly to zero.

Every remaining centered support contains zero in its convex hull, so its shrinking diameter bounds its absolute values as well. Equation (3.9) then implies that their total original variance tends to zero: their scaled laws converge uniformly to δ_0 , which is separated from $\text{Law}(\rho_E Z_E)$. The variances of the two-cluster coordinates in scaled units are bounded above and below by fixed positive constants. Their total original variance tends to one, and so $m_j \asymp \ell_j^{-2}$. Finally,

$$\sum_r \text{Var } Z_{r,j} = \ell_j^{-2} \sum_{\text{remaining } r} \text{Var } X_{r,j} = o(\ell_j^{-2}) = o(m_j).$$

□

5 An exact comparison near a common diameter

The following local theorem includes the additional coordinates in Proposition 4.3. Its proof adapts the two regimes of He and Cheng [3, Section 3]. Conditioning on a Bernoulli count now leaves different label probabilities and different gaps; we prove the conditional estimates explicitly.

Theorem 5.1 (Local comparison). *For each $\alpha \in (0, 1/2)$ there exist $\eta > 0$ and an integer N with the following property. Suppose $m \geq N$, $h > 0$, and*

$$X_i = h_i(B_i - p_i) + U_i, \quad 1 \leq i \leq m,$$

where the pairs (B_i, U_i) are independent, $B_i \sim \text{Bern}(p_i)$, $p_i \in [\alpha, 1 - \alpha]$, $\mathbb{E}(U_i \mid B_i) = 0$, and

$$|h_i/h - 1| \leq \eta, \quad |U_i| \leq \eta h.$$

There may also be any finite family of centered variables Z_r , mutually independent and independent of all the pairs, satisfying

$$|Z_r| \leq \eta h, \quad \sum_r \text{Var } Z_r \leq \eta m h^2.$$

Then the entire array satisfies

$$\mathcal{R}(X_1, \dots, X_m, Z_1, \dots) \leq C_E.$$

We prove this by considering any sequence with $m \rightarrow \infty$, $p_i \in [\alpha, 1 - \alpha]$, gap and individual residual bounds tending to zero, and total extra variance $o(mh^2)$. If the theorem were false, taking $\eta \downarrow 0$ and $N \rightarrow \infty$ would give a violating sequence of exactly this kind. Within a row put

$$v_i = p_i(1 - p_i), \quad \tau_i = p_i^2 + (1 - p_i)^2, \quad V = \sum_i v_i, \quad T_0 = \sum_i v_i \tau_i, \quad \mu = \sum_i p_i.$$

Scale the whole array by the variance-weighted mean gap $\bar{h} = V^{-1} \sum_i v_i h_i$. In the new units write

$$\begin{aligned} h_i &= 1 + a_i, & \sum_i v_i a_i &= 0, & K &= \sum_i B_i, \\ A &= \sum_i a_i(B_i - p_i), & U &= \sum_i U_i + \sum_r Z_r, & W &= A + U. \end{aligned} \tag{5.1}$$

The uncentered sum $S = K + W$ has mean μ . Define

$$\Lambda_A = \sum_i v_i a_i^2, \quad \Lambda_U = \sum_i \mathbb{E} U_i^2 + \sum_r \mathbb{E} Z_r^2, \quad \Lambda = \Lambda_A + \Lambda_U, \quad B^2 = V + \Lambda. \quad (5.2)$$

Conditional centering and (5.1) give exactly $\text{Cov}(K, W) = 0$ and $\text{Var } S = B^2$. Along the sequence,

$$V \asymp_\alpha m, \quad T_0 \asymp_\alpha m, \quad \max_i |a_i| \rightarrow 0, \quad \varepsilon := \max\{\max_i \|U_i\|_\infty, \max_r \|Z_r\|_\infty\} \rightarrow 0, \quad \Lambda = o(m).$$

Empty maxima are zero.

Let T_X be the actual sum of absolute third moments of all the centered coordinates, and let κ be their signed third-moment sum. We will use

$$|T_X - T_0| \leq C_\alpha \{\sqrt{m\Lambda_A} + \Lambda\}, \quad |\kappa - \kappa_0| \leq C_\alpha \{\sqrt{m\Lambda_A} + \Lambda\}, \quad \kappa_0 = \sum_i v_i(1 - 2p_i). \quad (5.3)$$

To prove these bounds, for large rows the signs of $(1 + a_i)(B_i - p_i) + U_i$ on the two clusters agree with those of $B_i - p_i$. Expand the signed and absolute cubes. The terms linear in U_i vanish conditionally; the others are bounded by $C_\alpha \mathbb{E} U_i^2$. The pure two-point absolute third moment is $(1 + a_i)^3 v_i \tau_i$, whose difference from $v_i \tau_i$ is bounded by $C_\alpha v_i(|a_i| + a_i^2)$. Cauchy–Schwarz gives $\sum_i v_i |a_i| \leq \sqrt{V\Lambda_A}$. The signed moment has the same bounds, and the extra variables contribute at most $\varepsilon \sum_r \mathbb{E} Z_r^2$. This proves (5.3). In particular $T_X = T_0 + o(m)$ and $B^2 = V + o(m)$.

5.1 The count and its conditional moments

Lemma 5.2. *Uniformly in $p_i \in [\alpha, 1 - \alpha]$ and integers k ,*

$$b_k := \mathbb{P}(K = k) = \frac{\phi((k - \mu)/\sqrt{V})}{\sqrt{V}} + O_\alpha(m^{-1}), \quad \sup_k b_k \leq C_\alpha m^{-1/2}. \quad (5.4)$$

For each fixed $M < \infty$, uniformly when $|k - \mu| \leq M\sqrt{m}$,

$$m_k := \mathbb{E}(A \mid K = k) = O_{\alpha, M}(\sqrt{\Lambda_A}/\sqrt{m}), \quad (5.5)$$

$$\text{Var}(A \mid K = k) = \Lambda_A \{1 + O_{\alpha, M}(m^{-1/2})\}, \quad (5.6)$$

$$\mathbb{E}[(A - m_k)^4 \mid K = k] \leq C_{\alpha, M} \Lambda_A^2. \quad (5.7)$$

Every error in these last three assertions has the displayed homogeneous dependence on the weight vector (a_i) .

Proof. Fourier inversion on $[-\pi, \pi]$ applies to the count. For its centered characteristic function F ,

$$|F(t)| \leq \exp\{-2V \sin^2(t/2)\} \leq e^{-c_\alpha m t^2}.$$

On a sufficiently small fixed interval, $\log F(t) = -Vt^2/2 + O_\alpha(m|t|^3)$. The real remainder can be absorbed in the quadratic term, and hence the difference from $e^{-Vt^2/2}$ has integral $O_\alpha(m \int_{\mathbb{R}} |t|^3 e^{-cmt^2} dt) = O_\alpha(m^{-1})$. Off that interval both the count transform and Gaussian integral have exponentially small tails. Inversion proves the first formula in (5.4), uniformly because its translating phase has modulus one. Integrating $|F|$ gives the second. The same statements hold after deleting any fixed number of coordinates. In particular,

$$\frac{\mathbb{P}(K - B_i = k - b)}{\mathbb{P}(K = k)} = 1 + O_{\alpha, M}(m^{-1/2}), \quad b = 0, 1, \quad |k - \mu| \leq M\sqrt{m}. \quad (5.8)$$

Indeed the means, variances and thresholds in numerator and denominator differ by bounded amounts, their central Gaussian masses have relative error $O_{\alpha,M}(m^{-1/2})$, and the absolute error in (5.4) has that same relative order in this region.

For a central k , choose the unique real θ such that

$$r_i = \frac{p_i e^\theta}{1 - p_i + p_i e^\theta}, \quad \sum_i r_i = k.$$

Then $\theta = O_{\alpha,M}(m^{-1/2})$, since the derivative of the last sum is of order m on a fixed neighborhood of zero. All r_i remain in one compact subinterval of $(0, 1)$. Tilting by $e^{\theta K}$ leaves the conditional law given $K = k$ unchanged. With expectations under this tilted product law denoted by $\mathbb{E}_\theta$, set

$$w_i = r_i(1 - r_i), \quad V_\theta = \sum_i w_i, \quad c_\theta = \frac{\sum_i w_i a_i}{V_\theta}, \quad b'_i = a_i - c_\theta.$$

On $K = k$ we have

$$A = \sum_i a_i(r_i - p_i) + L, \quad L = \sum_i b'_i(B_i - r_i), \quad \sum_i w_i b'_i = 0. \quad (5.9)$$

The expansions $r_i - p_i = \theta v_i + O_\alpha(\theta^2 v_i)$ and $w_i = v_i\{1 + O_\alpha(|\theta|)\}$ give

$$\left| \sum_i a_i(r_i - p_i) \right| \leq C_{\alpha,M} \sqrt{\Lambda_A} / \sqrt{m}, \quad (5.10)$$

$$Q_\theta := \sum_i w_i (b'_i)^2 = \Lambda_A \{1 + O_{\alpha,M}(m^{-1/2})\}, \quad Q := \sum_i (b'_i)^2 \asymp_\alpha \Lambda_A. \quad (5.11)$$

For the first estimate the linear term is exactly zero by (5.1). For the second, expand $Q_\theta = \sum_i w_i a_i^2 - (\sum_i w_i a_i)^2 / V_\theta$; the subtracted term is $O_{\alpha,M}(\Lambda_A/m)$. If $\Lambda_A = 0$, all the conditional assertions are immediate. Assume henceforth $Q > 0$.

Here are the required Fourier moment estimates. Put $Z'_i = B_i - r_i$, $f_i(t) = \mathbb{E}_\theta e^{itZ'_i}$, and $F_\theta(t) = \prod_i f_i(t)$. For small fixed $|t|$, these factors are bounded away from zero. Let $c_{i,r}(t)$ be the r th derivative at $u = 0$ of

$$\log \frac{\mathbb{E}_\theta e^{(it+u)Z'_i}}{\mathbb{E}_\theta e^{itZ'_i}}.$$

Uniformly on that interval,

$$c_{i,1}(t) = iw_i t + O_\alpha(t^2), \quad c_{i,2}(t) = w_i + O_\alpha(|t|), \quad |c_{i,3}(t)| + |c_{i,4}(t)| \leq C_\alpha.$$

Define $M_r(t) = \sum_i (b'_i)^r c_{i,r}(t)$. Cancellation in (5.9), together with $\sum_i |b'_i| \leq \sqrt{mQ}$ and $\sum_i |b'_i|^r \leq Q^{r/2}$ for $r \geq 2$, gives

$$\begin{aligned} |M_1(t)| &\leq C_\alpha \sqrt{mQ} t^2, & M_2(t) &= Q_\theta + O_\alpha(Q|t|), \\ |M_3(t)| &\leq C_\alpha Q^{3/2}, & |M_4(t)| &\leq C_\alpha Q^2. \end{aligned} \quad (5.12)$$

Differentiating the exponential of the sum of logarithms shows that the Fourier numerators for L, L^2, L^4 are

$$\begin{aligned} F_\theta M_1, & \quad F_\theta(M_1^2 + M_2), \\ F_\theta(M_1^4 + 6M_1^2 M_2 + 3M_2^2 + 4M_1 M_3 + M_4). \end{aligned}$$

Their integrals over $[-\pi, \pi]$, divided by 2π , are $\mathbb{E}_\theta[L^r \mathbf{1}_{\{K=k\}}]$, since $\sum_i r_i = k$ is an integer. On the central frequency interval use $|F_\theta(t)| \leq e^{-c_\alpha m t^2}$ and

$$\int_{\mathbb{R}} |t|^r e^{-c_\alpha m t^2} dt = C_{c,r} m^{-(r+1)/2}.$$

After division by $\mathbb{P}_\theta(K = k) \asymp_\alpha m^{-1/2}$, the first numerator gives $O(\sqrt{Q}/\sqrt{m})$. In the second, the constant $Q_\theta F_\theta$, integrated over the full frequency interval, gives Q_θ exactly; its omitted central-interval tail is exponentially small. The $O(Q|t|)$ term gives $O(Q/\sqrt{m})$, and M_1^2 gives $O(Q/m)$. For the fourth numerator, the five displayed terms give, respectively,

$$O(Q^2/m^2), \quad O(Q^2/m), \quad O(Q^2), \quad O(Q^2/\sqrt{m}), \quad O(Q^2).$$

Away from the central interval, expand $\mathbb{E}_\theta[L^r e^{it(K-k)}]$ before factoring. Each term contains at most r chosen coordinate factors and retains exponential decay from the remaining factors. Its absolute integral is at most

$$C_{\alpha,r} e^{-c_\alpha m} \left(\sum_i |b'_i| \right)^r \leq C_{\alpha,r} e^{-c_\alpha m} (mQ)^{r/2} \quad (r = 1, 2, 4).$$

This is absorbed in the bounds just obtained after conditioning; in particular it contains no additive error independent of Q . We have proved

$$\mathbb{E}_\theta(L \mid K = k) = O(\sqrt{Q}/\sqrt{m}), \quad \mathbb{E}_\theta(L^2 \mid K = k) = Q_\theta + O(Q/\sqrt{m}), \quad \mathbb{E}_\theta(L^4 \mid K = k) \leq CQ^2.$$

Subtracting the conditional mean, and using (5.10)–(5.11), proves (5.5)–(5.7). $\square$

Lemma 5.3. *For every fixed M , uniformly when $|k - \mu| \leq M\sqrt{m}$,*

$$\mathbb{E}(W \mid K = k) = m_k, \quad \text{Var}(W \mid K = k) = \Lambda \{1 + O_{\alpha,M}(m^{-1/2})\}.$$

Put

$$D = \Lambda^2 + \varepsilon^2 \Lambda_U, \quad H = \max \left\{ \sqrt{\Lambda_A}, \frac{\Lambda}{\sqrt{\Lambda + \varepsilon^2}} \right\} \quad (\Lambda > 0).$$

Then

$$\mathbb{E}[(W - m_k)^4 \mid K = k] \leq C_{\alpha,M} D, \quad \mathbb{E}(|W - m_k| \mid K = k) \geq c_{\alpha,M} H, \quad |m_k| \leq C_{\alpha,M} H / \sqrt{m}. \quad (5.13)$$

Also $\mathbb{E}W^4 \leq C_\alpha D$ without conditioning.

Proof. Write $s_{ib} = \mathbb{E}(U_i^2 \mid B_i = b)$. Given the full label vector $\mathbf{b}$, the summands in U are independent and centered, and their variance is $T_{\mathbf{b}} = \sum_i s_{i,b_i} + \sum_r \mathbb{E}Z_r^2$. Since both label probabilities are at least α , $T_{\mathbf{b}} \leq \Lambda_U/\alpha$. Equation (5.8) gives the relative estimate

$$\mathbb{E}(T_{\mathbf{b}} \mid K = k) = \Lambda_U \{1 + O_{\alpha,M}(m^{-1/2})\}.$$

For independent centered variables, expansion of the fourth power gives a fourth moment at most the sum of fourth moments plus three times the squared variance. Thus

$$\mathbb{E}(U^4 \mid \mathbf{B}) \leq 3T_{\mathbf{b}}^2 + \varepsilon^2 T_{\mathbf{b}},$$

and averaging preserves the bound $C_\alpha(\Lambda_U^2 + \varepsilon^2\Lambda_U)$. The conditional covariance between A and U is zero, because A is determined by the labels and $\mathbb{E}(U \mid \mathbf{B}) = 0$. Lemma 5.2 proves the mean, variance and fourth-moment claims for W .

Interpolation in the form $\mathbb{E}|Y| \geq (\mathbb{E}Y^2)^{3/2}/(\mathbb{E}Y^4)^{1/2}$ gives

$$\mathbb{E}(|W - m_k| \mid K = k) \geq c_{\alpha,M} \frac{\Lambda}{\sqrt{\Lambda + \varepsilon^2}}.$$

There is also a bound from the gap residual alone. Conditional Jensen, first given the full labels, gives

$$\mathbb{E}(|A + U - m_k| \mid K = k) \geq \mathbb{E}(|A - m_k| \mid K = k) \geq c_{\alpha,M} \sqrt{\Lambda_A},$$

using (5.6)–(5.7). These prove the bound by H , and (5.5) proves the drift bound. Finally, $\max_i a_i^2 \leq \Lambda_A/[\alpha(1 - \alpha)]$, so direct fourth moment expansion gives $\mathbb{E}A^4 \leq C_\alpha\Lambda_A^2$. The preceding full-label bound for U then gives $\mathbb{E}W^4 \leq C_\alpha D$. $\square$

5.2 Uniform jitter before standardization

Lemma 5.4. *Along the sequences specified after Theorem 5.1, let J be the CDF of $S + V_1$, where V_1 is independent and uniform on $[-1/2, 1/2]$. Uniformly over $x \in \mathbb{R}$, with $z = (x - \mu)/B$,*

$$J(x) = \Phi(z) + \frac{\kappa}{6B^3}(1 - z^2)\phi(z) + o(m^{-1/2}). \quad (5.14)$$

Both signed discrepancies of S , multiplied by B^3/T_X , are therefore bounded above by

$$\phi(z) \left\{ \frac{B^2}{2T_X} + \frac{|\kappa|}{6T_X}|1 - z^2| \right\} + o(1). \quad (5.15)$$

Proof. Set $d_m = T_X/B^2$. By (5.3), d_m lies eventually in a fixed compact subinterval of $(0, \infty)$, and each subsequence has a further one with $d_m \rightarrow d > 0$. Apply Lemma 3.1 to the centered array divided by B . Its Lyapunov ratio is $d_m/B \asymp_\alpha m^{-1/2}$ and each coordinate is bounded by a fixed multiple of that ratio. In the measure (3.1), the variables before symmetrization are the raw coordinates divided by d_m .

The total squared difference between two-cluster coordinates and $B_i - p_i$, divided by B^2 , is $O(\Lambda/m) = o(1)$. The additional coordinates contribute $\sum_r \text{Var } Z_r/B^2 = o(1)$ to this variance-weighted measure. For the Bernoulli coordinates, the nonzero symmetrized differences are 1 and -1 , each with probability v_i ; their difference variance is $2v_i$. Thus the limit of (3.1) is

$$\frac{1}{2}\delta_{-1/d} + \frac{1}{2}\delta_{1/d}.$$

Thus the jitter parameter in Lemma 3.1 is $h = 1/d$. That lemma gives (5.14) with raw jitter width $d_m/d \rightarrow 1$. Couple this jitter with width one. Their difference is $o(1)$ in raw units, or $o(B^{-1})$ after standardization. Bounding the two CDFs by shifted versions of one another, using the bounded derivative of the smooth comparison function, preserves the $o(m^{-1/2})$ error. The subsequence argument proves the asserted expansion along the original sequence. Shift the jittered CDF by $\pm 1/2$ to obtain (5.15), also for the strict lower branch. $\square$

5.3 Comparison when the residual variance tends to zero

We need a one-sided estimate that keeps the size of the residual in its error. The following slightly enlarged version of He and Cheng [3, Lemma 3.2] follows from their clipping proof; we include the enlargement to accommodate the conditional mean.

Lemma 5.5 (One-sided loss). *There are absolute constants $c_0, C, \kappa_* > 0$ such that, if $\mathbb{E}Y = 0$, $\mathbb{E}Y^4 \leq \kappa_*$, and $f \in C^1([-7/10, 7/10])$ satisfies $f(0) = 0$ and $3/4 \leq f' \leq 5/4$, then for $|u| \leq 3/5$,*

$$\mathbb{P}(Y > u) + f(u) \geq c_0 \mathbb{E}|Y| - CEY^4, \quad \mathbb{P}(Y < u) - f(u) \geq c_0 \mathbb{E}|Y| - CEY^4. \quad (5.16)$$

Proof. First let V be centered with $|V| \leq a = 1/10$ and $|u| \leq 13/20$. Writing $V_+ = \max(V, 0)$, for $u \geq 0$ the bounds $\mathbb{E}V_+ \leq u + a\mathbb{P}(V > u)$ and $f(u) \geq 3u/4$ give $\mathbb{P}(V > u) + f(u) \geq 3\mathbb{E}|V|/8$. If $u = -r$ with $0 < r \leq a$, put $t = \mathbb{P}(V > -r)$. Centering implies $0 \leq at - r(1 - t)$, so $r \leq 2at$. Also $\mathbb{E}|V| = 2\mathbb{E}V_+ \leq 2at$. Hence $t + f(-r) \geq t - 5r/4 \geq 3t/4 \geq 3\mathbb{E}|V|/8$. For $a < r \leq 13/20$ the probability is one, and $1 + f(-r) \geq 3/16 \geq 3\mathbb{E}|V|/8$.

For general Y , clip it at $a_0 = 1/20$: put $T = \max(-a_0, \min(Y, a_0))$, $b = \mathbb{E}T$, and $V = T - b$. Writing $M_4 = \mathbb{E}Y^4$,

$$|b| \leq a_0^{-3} M_4, \quad \mathbb{P}(T \neq Y) \leq a_0^{-4} M_4, \quad \mathbb{E}|V| \geq \mathbb{E}|Y| - 2a_0^{-3} M_4.$$

Choose κ_* so that $|b| \leq 1/20$. Then $|V| \leq 1/10$ and $|u - b| \leq 13/20$. Apply the preceding estimate at $u - b$, using

$$\mathbb{P}(Y > u) \geq \mathbb{P}(V > u - b) - \mathbb{P}(T \neq Y), \quad |f(u) - f(u - b)| \leq 5|b|/4.$$

This proves the first inequality. Apply it to $-Y$ and $-f(-\cdot)$ to prove the second. The strict inequalities in the events throughout also cover atoms at u . $\square$

Proposition 5.6. *Along any subsequence of the local arrays for which $\Lambda \rightarrow 0$,*

$$\mathcal{R}(X_1, \dots, X_m, Z_1, \dots) \leq \mathcal{R}(B_1 - p_1, \dots, B_m - p_m) \leq C_E$$

for all sufficiently large rows. The first inequality is strict when $\Lambda > 0$.

Proof. If $\Lambda = 0$, all residuals vanish and (1.4) applies directly. Consider the rows with $\Lambda > 0$. Write $R_0 = \mathcal{R}(B_1 - p_1, \dots, B_m - p_m)$. We compare both signed branches at an arbitrary threshold $x = k + u$, where k is a nearest integer and $|u| \leq 1/2$. The exact count decompositions are

$$F_S(k + u) = F_K(k) - b_k \mathbb{P}(W > u \mid K = k) + e_{k,u}^+, \quad (5.17)$$

$$F_S^-(k + u) = F_K(k - 1) + b_k \mathbb{P}(W < u \mid K = k) + e_{k,u}^-. \quad (5.18)$$

For either remainder, the absolute value is bounded by

$$\sum_{j \neq k} b_j \mathbb{P}\{|W| \geq |j - k| - 1/2 \mid K = j\}.$$

Fix M . For $|k - \mu| \leq M\sqrt{m}$, split this sum at $|j - k| \leq \sqrt{m}$. For the nearby count values, the conditional fourth moment is at most $C_{\alpha,M}D$ by Lemma 5.3, including the fourth power of the mean. Use $b_j \leq C_{\alpha}/\sqrt{m}$ and the convergent series $\sum_{r \neq 0} (|r| - 1/2)^{-4}$. For the remote count values, the denominator is at least cm^2 , and summing the conditional fourth moments with their probabilities gives $\mathbb{E}W^4 \leq C_{\alpha}D$. Therefore, uniformly in central k and $|u| \leq 1/2$,

$$|e_{k,u}^{\pm}| \leq C_{\alpha,M}D/\sqrt{m}. \quad (5.19)$$

Put

$$G_k(u) = \Phi((k - \mu + u)/B), \quad f_k(u) = \{G_k(u) - G_k(0)\}/b_k.$$

Equation (5.4) implies $3/4 \leq f'_k(u) \leq 5/4$ on $[-1, 1]$ for all sufficiently large rows, uniformly in central k . Indeed, $f'_k(u) = \phi((k - \mu + u)/B)/(Bb_k)$, and $B^2 = V + \Lambda$ with $\Lambda \rightarrow 0$. This use of the relative count estimate introduces no additive discrepancy error.

Let $m_k = \mathbb{E}(W \mid K = k)$ and $Y = W - m_k$ under this conditional law. For large rows $|m_k| \leq 1/20$ and $\mathbb{E}(Y^4 \mid K = k) \leq \kappa_*$. Apply Lemma 5.5 to $\tilde{f}(v) = f_k(v + m_k) - f_k(m_k)$ at $v = u - m_k$. The derivative bound on $[-1, 1]$ covers all its evaluations. By Lemma 5.3, both losses satisfy

$$\begin{aligned} \mathbb{P}(W > u \mid K = k) + f_k(u) &\geq c_{\alpha, M}H - C_{\alpha, M}(D + |m_k|), \\ \mathbb{P}(W < u \mid K = k) - f_k(u) &\geq c_{\alpha, M}H - C_{\alpha, M}(D + |m_k|). \end{aligned} \quad (5.20)$$

Substitute (5.20) into (5.17)–(5.18), subtract the normal CDF, and multiply by $B^3/T_X \asymp_\alpha \sqrt{m}$. The product $(B^3/T_X)b_k$ is bounded above and below by positive constants on this central region. Changing the normal variance from V to $V + \Lambda$ costs at most $C_\alpha \Lambda/m$ before this multiplication, because $\sup_z |z\phi(z)| < \infty$. Changing the prefactor from $V^{3/2}/T_0$ to B^3/T_X costs at most

$$C_\alpha \{\sqrt{\Lambda_A}/\sqrt{m} + \Lambda/m\}$$

after normalization: use (5.3) and the bound $R_0 \leq C_U$. Including (5.19) and the conditional drift, either normalized central branch is at most

$$R_0 - c_{\alpha, M}H + C_{\alpha, M} \left\{ D + \frac{\sqrt{\Lambda_A} + \Lambda}{\sqrt{m}} \right\}. \quad (5.21)$$

All these errors are absorbable relative to H , since

$$\frac{D}{H} \leq (\Lambda + \varepsilon^2)^{3/2}, \quad \frac{\sqrt{\Lambda_A}}{H} \leq 1, \quad \frac{\Lambda}{H} \leq \sqrt{\Lambda + \varepsilon^2}. \quad (5.22)$$

For sufficiently large rows, (5.21) is therefore at most $R_0 - c_{\alpha, M}H/2$. This conclusion holds for every positive Λ , however small its value relative to m .

It remains to restrict the threshold to a fixed central region. A jump of size b_k gives a signed discrepancy on one of its two sides of at least $b_k/2$. Choose k nearest to μ in (5.4); since $T_0 \leq V$,

$$R_0 \geq \frac{V^{3/2}}{2T_0} \max_k b_k \geq \phi(0)/4$$

for large m . Meanwhile (5.15) is bounded by $C_\alpha(1 + z^2)\phi(z) + o(1)$, uniformly over all thresholds. Choose a fixed large radius so that this is below $\phi(0)/8$ outside the radius for large rows. Since $B \asymp_\alpha \sqrt{m}$, choose M to contain the remaining thresholds and their nearest integers. The far branches are at most $R_0 - \phi(0)/8$. Together with the central comparison this proves the strict inequality for $\Lambda > 0$. This is the exact-loss comparison of He and Cheng [3, Lemma 3.3], with the conditional estimates above in place of iid conditional symmetry. $\square$

5.4 Positive mass in intervals of fixed length

We next treat subsequences for which $\Lambda \geq \Lambda_* > 0$. The variance may grow, but remains $o(m)$.

Lemma 5.7. Fix $\Lambda_* > 0$, $R < \infty$, and real numbers $a < b$. Along a local sequence with $\Lambda \geq \Lambda_*$ there is $c_* > 0$, depending on these fixed parameters and α , such that eventually

$$\inf_{|x-\mu| \leq R\sqrt{m}} \mathbb{P}\{S \in (x+a, x+b)\} \geq c_* m^{-1/2}. \quad (5.23)$$

Proof. For each x in the indicated region choose all integers k with $|k-x| \leq r$, where $r = \max(1, \sqrt{\Lambda})$. There are at least $\sqrt{\Lambda}$ such integers. Since $\Lambda = o(m)$, they lie in a fixed enlarged central region for large rows, and each satisfies $b_k \geq c_{\alpha,R}/\sqrt{m}$. Tilt the labels to mean k and use the projection from the proof of Lemma 5.2. Conditional on $K = k$,

$$W = m_k^0 + R_k, \quad m_k^0 = \sum_i a_i(r_i - p_i), \quad R_k = \sum_i \{b'_i(B_i - r_i) + U_i\} + \sum_r Z_r.$$

Under the tilted independent law,

$$\mathbb{E}_\theta R_k = 0, \quad \text{Cov}_\theta(K, R_k) = 0, \quad L_k := \text{Var}_\theta R_k = \Lambda\{1 + O_{\alpha,R}(m^{-1/2})\}, \quad |m_k^0| \leq C_{\alpha,R}\sqrt{\Lambda_A}/\sqrt{m}. \quad (5.24)$$

The variance comparison follows from (5.11) and the relative $O(m^{-1/2})$ change in each label probability; the extra variables are unchanged. Every individual summand of R_k is bounded by a number $\delta_m \rightarrow 0$, uniformly over these k .

We prove a conditional local estimate for a fixed continuous integrable function g with integrable Fourier transform supported in $[-T, T]$:

$$\sup_{y \in \mathbb{R}} \left| \mathbb{E}_\theta[g(R_k - y) \mid K = k] - \mathbb{E}g(\sqrt{L_k} Z - y) \right| = o(L_k^{-1/2}), \quad Z \sim N(0, 1). \quad (5.25)$$

All errors here are uniform over the selected values of k . Let $\chi_k(s, t) = \mathbb{E}_\theta e^{is(K-k) + itR_k}$. For a sufficiently small fixed $\delta > 0$, symmetrization of each independent pair gives, for large rows,

$$|\chi_k(s, t)| \leq \exp\{-c_\alpha(V_\theta s^2 + L_k t^2)\}, \quad |s| \leq \delta, \quad |t| \leq T. \quad (5.26)$$

In detail, each pair phase has diameter at most $|s| + 2T\delta_m$. For this diameter small enough, $1 - \cos u \geq cu^2$ bounds its characteristic-function modulus by the exponential of minus a constant times its phase variance. The sum of those variances is exactly $V_\theta s^2 + L_k t^2$, by the zero covariance in (5.24). Extra coordinates have count component zero and contribute their actual variance to L_k . For $\delta \leq |s| \leq \pi$, each factor corresponding to a pair (B_i, U_i) differs from $\mathbb{E}_\theta e^{is(B_i - r_i)}$ by at most $T\delta_m$. The latter moduli are bounded away from one uniformly over this compact range of s and the allowed r_i . The product is therefore $O(e^{-cm})$; additional factors have modulus at most one.

For bounded u, v , at $s = u/\sqrt{V_\theta}$ and $t = v/\sqrt{L_k}$ the sum of absolute third phase moments is at most

$$C_\alpha \left\{ \frac{|u|^3}{\sqrt{m}} + \frac{\delta_m |v|^3}{\sqrt{L_k}} \right\} = o(1),$$

because $L_k \geq \Lambda_*/2$ eventually. The logarithmic characteristic-function expansion hence gives $\chi_k(s, t) - e^{-V_\theta s^2/2 - L_k t^2/2} \rightarrow 0$ uniformly on bounded scaled frequency sets. The Gaussian majorant (5.26) makes its integrated absolute difference on $[-\delta, \delta] \times [-T, T]$ equal to $o((V_\theta L_k)^{-1/2})$: after the change of variables, first restrict to a bounded rectangle and then use Gaussian tails. The other count frequencies are exponentially small at this scale, since $L_k = o(m)$.

Fourier inversion in the integer count coordinate and in the residual coordinate gives

$$\mathbb{E}_\theta[g(R_k - y)\mathbf{1}_{\{K=k\}}] = \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \int_{-T}^T \hat{g}(t) e^{ity} \chi_k(s, -t) dt ds.$$

The translating factor has modulus one and $|\hat{g}| \leq \|g\|_1$. The preceding integrated estimate therefore applies uniformly in y . Replacing χ_k by the Gaussian product gives $(2\pi V_\theta)^{-1/2} \mathbb{E}g(\sqrt{L_k}Z - y)$, up to an exponentially small count tail. Divide by

$$\mathbb{P}_\theta(K = k) = (2\pi V_\theta)^{-1/2} \{1 + O_\alpha(m^{-1/2})\},$$

from (5.4). The denominator replacement is also $o(L_k^{-1/2})$, since $|\mathbb{E}g(\sqrt{L_k}Z - y)| \leq \phi(0)\|g\|_1/\sqrt{L_k}$. This proves (5.25), including when $L_k \rightarrow \infty$.

To obtain an interval probability, put $w_0 = (a + b)/2$ and $d = (b - a)/2$, and for a fixed large $\omega > 0$ define

$$g(w) = \left(1 - \frac{(w - w_0)^2}{d^2}\right) \left(\frac{\sin(\omega(w - w_0))}{\omega(w - w_0)}\right)^4.$$

Then $g \leq \mathbf{1}_{(a,b)}$, including at the endpoints, and $g \in L^1(\mathbb{R})$. The Fourier transform of the fourth power of sinc is a compactly supported cubic spline. Multiplication by the quadratic corresponds to two derivatives, so $\hat{g}$ is continuous, integrable and supported in $[-4\omega, 4\omega]$. For any fixed C_0 , choose ω sufficiently large that

$$\mathbb{E}g(\sqrt{\lambda}Z - y) \geq c_1/\sqrt{\lambda} \quad (\lambda \geq \Lambda_*/2, \quad |y| \leq C_0\sqrt{\lambda}). \quad (5.27)$$

Here is a direct lower bound. On $|w - w_0| \leq 1/\omega$ the positive part of g has integral at least c/ω . The Gaussian density there is bounded below by $c_{\Lambda_*, C_0, a, b}/\sqrt{\lambda}$. The negative part has integral at most $2/(d^3\omega^4)$, while the density is at most $\phi(0)/\sqrt{\lambda}$. Fixing ω large enough makes the positive contribution dominate and proves (5.27).

For each chosen k set $y = x - k - m_k^0$. By (5.24) and $|k - x| \leq r$, $|y| \leq C_{\Lambda_*, R}\sqrt{L_k}$. Use (5.25)–(5.27) to get

$$\mathbb{P}\{W \in (x + a - k, x + b - k) \mid K = k\} \geq c_2/\sqrt{\Lambda}.$$

Each chosen value of K has probability at least $c_{\alpha, R}/\sqrt{m}$, and there are at least $\sqrt{\Lambda}$ such values. Summing proves (5.23). $\square$

Proposition 5.8. *Along a local sequence with $\Lambda \geq \Lambda_* > 0$,*

$$\limsup \mathcal{R}(X_1, \dots, X_m, Z_1, \dots) < C_E.$$

Proof. The uniform-jitter identities, with strict events, are

$$\begin{aligned} J(x + 1/2) - F_S(x) &= \mathbb{E}[(x + 1 - S)\mathbf{1}_{\{x < S < x+1\}}], \\ F_S^-(x) - J(x - 1/2) &= \mathbb{E}[(S - x + 1)\mathbf{1}_{\{x-1 < S < x\}}]. \end{aligned}$$

Lemma 5.7, applied for example to the intervals $(x + 1/4, x + 1/2)$ and $(x - 1/2, x - 1/4)$, makes both quantities at least $c_R/\sqrt{m}$ in any fixed central region. Combine this with (5.14) and Taylor expansion at the shifts $\pm 1/2$. Multiplying by B^3/T_X , either central branch is at most

$$\frac{\phi(0)(3B^2 + |\kappa|)}{6T_X} - c'_R + o(1). \quad (5.28)$$

The Bernoulli case of (1.3), with span one, gives $3V + |\kappa_0| \leq c_E T_0$. Equations (5.3) and $\Lambda = o(m)$ show that the first term of (5.28) is at most $C_E + o(1)$. Outside a sufficiently large fixed standardized region, the ordinary envelope (5.15) is less than $C_E/2$ for large rows. Choose this region first, then its positive central loss c'_R . This proves the strict upper limit. The interval-loss step corresponds to He and Cheng [3, Lemmas 3.4–3.5]; the two-dimensional calculation above supplies the mass bound for unequal probabilities and gaps. $\square$

Proof of Theorem 5.1. If the theorem failed, select violating rows with $m_j \geq j$, gap and residual bounds at most $1/j$, and extra variance at most $m_j h^2/j$. After the common normalization (5.1), these have exactly the sequential hypotheses used above. If Λ_j failed to tend to zero, a subsequence would have $\Lambda_j \geq \Lambda_* > 0$, contradicting Proposition 5.8. Thus $\Lambda_j \rightarrow 0$. Proposition 5.6 and (1.4) then contradict the strict violations. $\square$

6 Completion of the proof

Proof of Theorem 1.1. If violations existed with arbitrarily small Lyapunov ratio, Lemma 2.1 would produce exact penalized maximizers with $R_j > C_E$ and $\ell_j \rightarrow 0$. Proposition 2.3 gives their common support bound. Proposition 3.2 and the support arguments of Section 4 put every sufficiently late row in the class of Theorem 5.1: use the fixed nominal gap $h = 1/\tau_E$, the probability interval from Proposition 4.3, and $m_j \asymp \ell_j^{-2} \rightarrow \infty$. The remaining coordinates have shrinking support and total scaled variance $o(m_j)$, as required. Theorem 5.1, which uses all actual variances and third absolute moments, therefore gives $R_j \leq C_E$, a contradiction. Some universal $\ell_0 > 0$ must exist. The Esseen iid sequence described in Section 1 proves optimality. $\square$

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